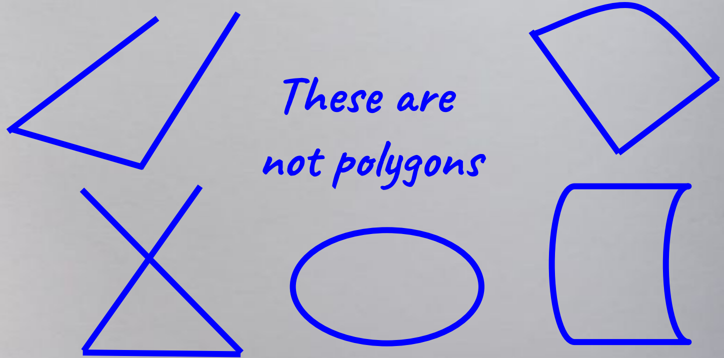
Geotopia University

Professor's Note

"Imagination is more important than knowledge. For knowledge is limited, whereas imagination embraces the entire world, stimulating progress, giving birth to evolution."

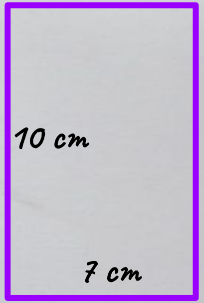
- Albert Einstein

Note #3 Polygons



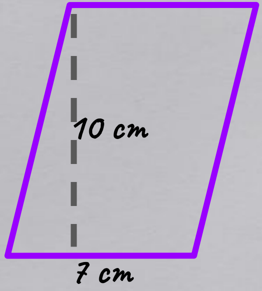
Area of Rectangle:
Length times width

$10 \times 7 = 70\ cm^2$



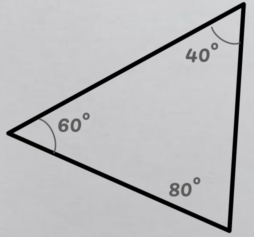
Area of Parallelogram:
Base times height

$10 \times 7 = 70\ cm^2$

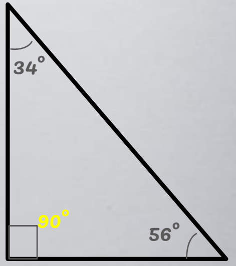


Note #8 Triangles

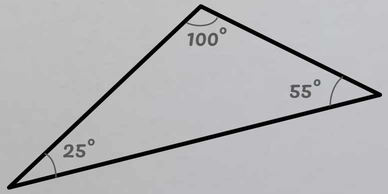
Acute Triangle



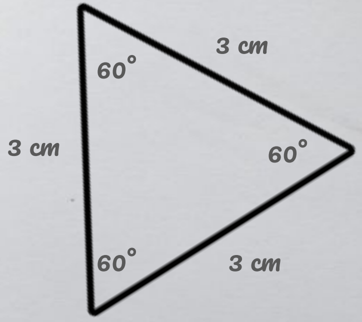
Right Triangle



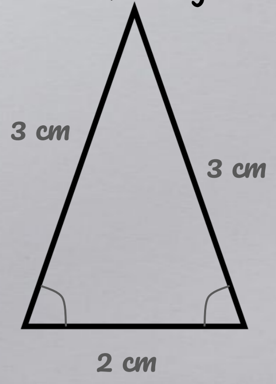
Obtuse Triangle



Equilateral Triangle

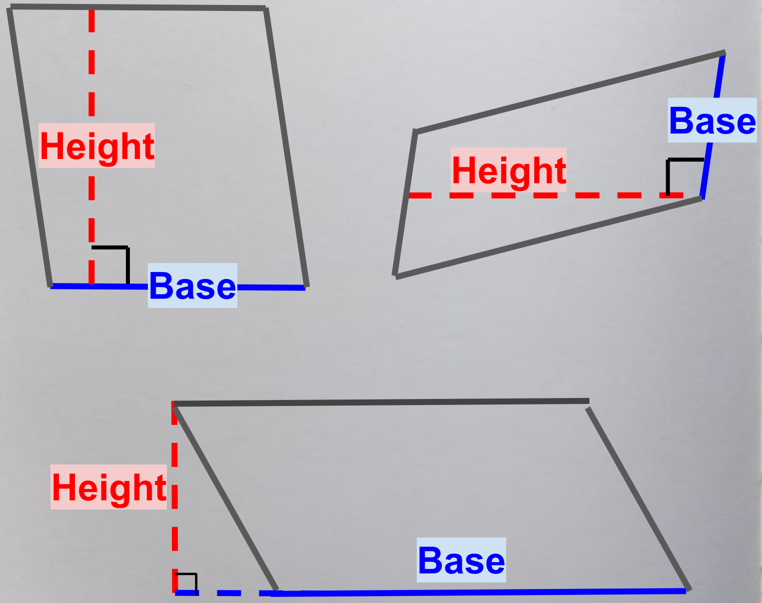


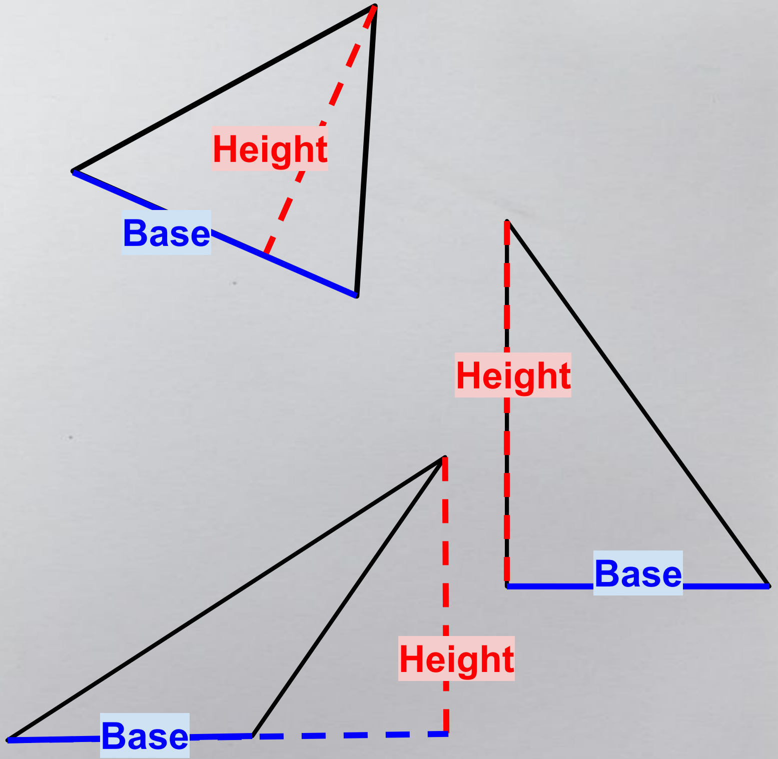
Isosceles Triangle



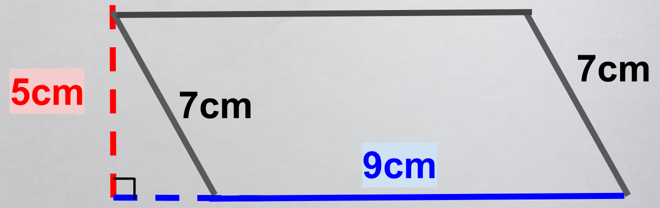
Note #13 Base and Height

Base and height always form a 90° angle! (right angle)



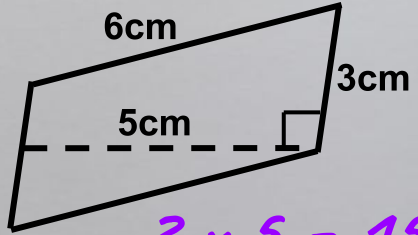


Note #15 Area

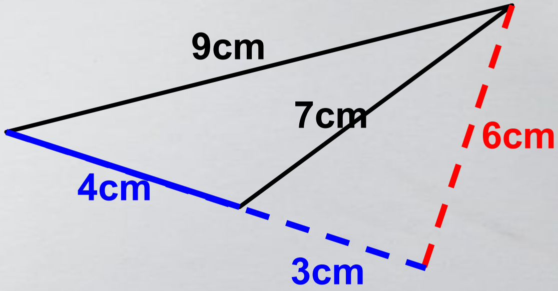


Area of Parallelogram = Base times height

$9 \times 5 = 45\ cm^2$

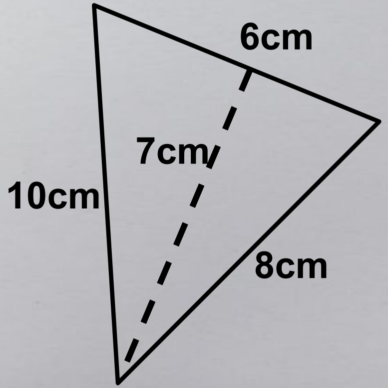


$3 \times 5 = 15\ cm^2$



Area of triangle = half of (Base times height)

$4 \times 6 \div 2 = 12\ cm^2$



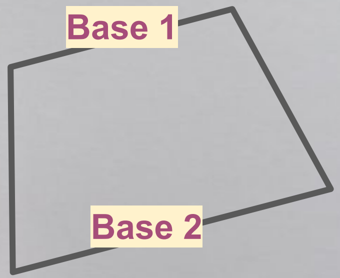
$6 \times 7 \div 2$

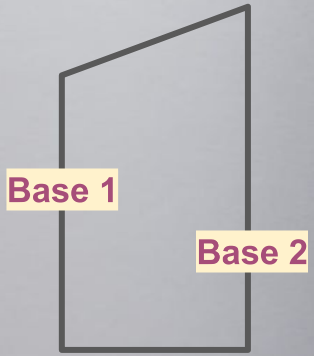
$= 21\ cm^2$

Note #20 Area of trapezoid

There are 2 bases in a trapezoid.
They are base 1 and base 2.
They are parallel to each other.

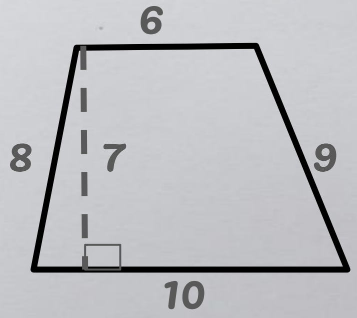
(Parallel lines are going to the same direction and never go across each other)





Area of trapezoid =

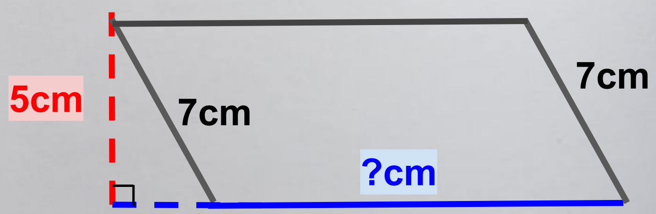
(Base 1 + Base 2) x Height ÷ 2



$(6 + 10) \times 7 \div 2 = 56\ cm^2$

Note #15 Finding missing length

The area is 45 cm^2

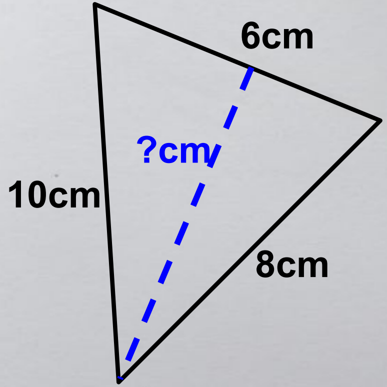


The missing base is

$45 \div 5 = 9$ cm

Area Height Base

The area is 12 cm^2



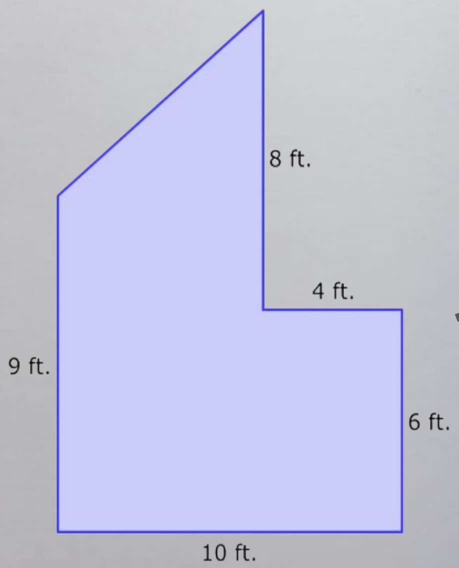
The missing height is

$12 \times 2 \div 6 = 4$ cm

Double the area Base Height

Note #23 Composite Figure

How can we find the area of this composite figure?



Cut!

Triangle A:

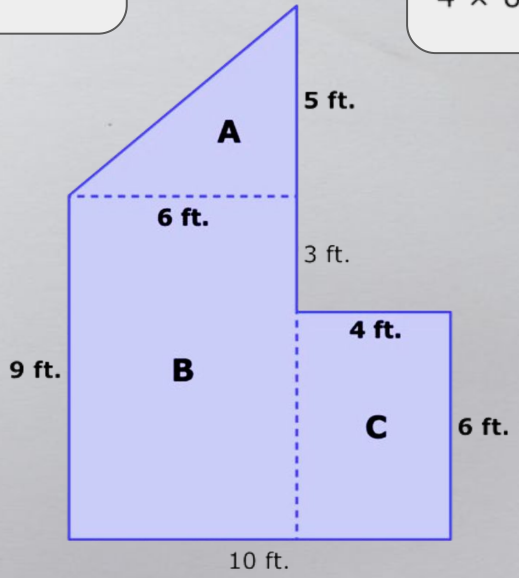
$\frac{1}{2} \times 6 \times 5 = 15$

Rectangle B:

$9 \times 6 = 54$

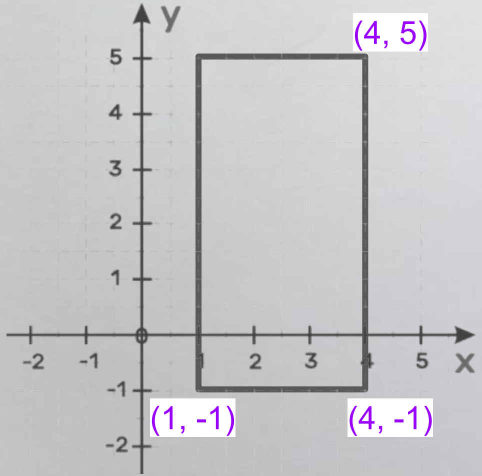
Rectangle C:

$4 \times 6 = 24$



$$15 + 54 + 24 = 93\ ft^2$$

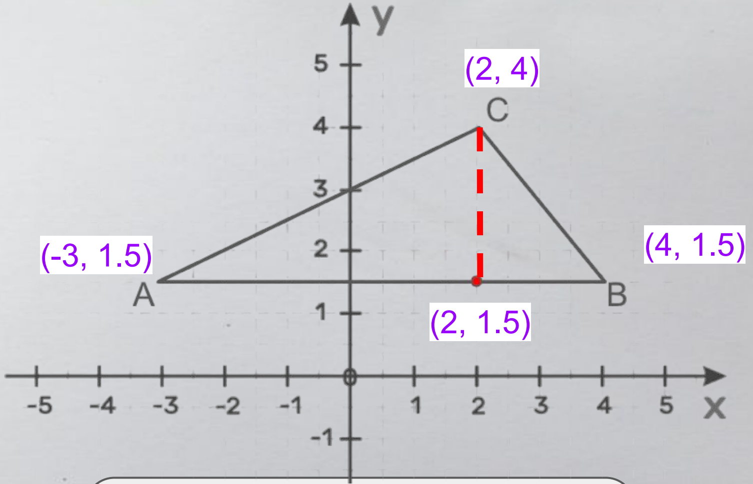
Note #29 Coordinate Plane



From (1, -1) to (4, -1)
The base is 3

From (4, 5) to (4, -1)
The base is 6

Area= 6 x 3 = 18



From (-3, 1.5) to (4, 1.5)
The base is 7

From (2, 4) to (2, 1.5)
The height is 3.5

Area= 3.5 x 7 ÷ 2 = 12.25

Note #32 Perfect Square and Perfect Cube

This is 4 because it is 2x2.

This is 9 because it is 3x3.

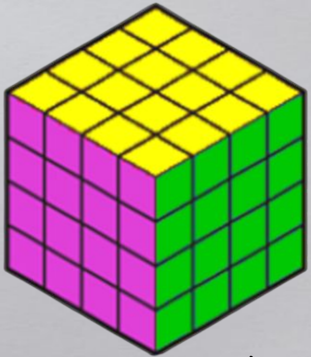
This is 16 because it is 4x4.

The number 4, 9, 16, 25, 36... are perfect squares!

1x1x1=1 cube

2x2x2=8 cubes

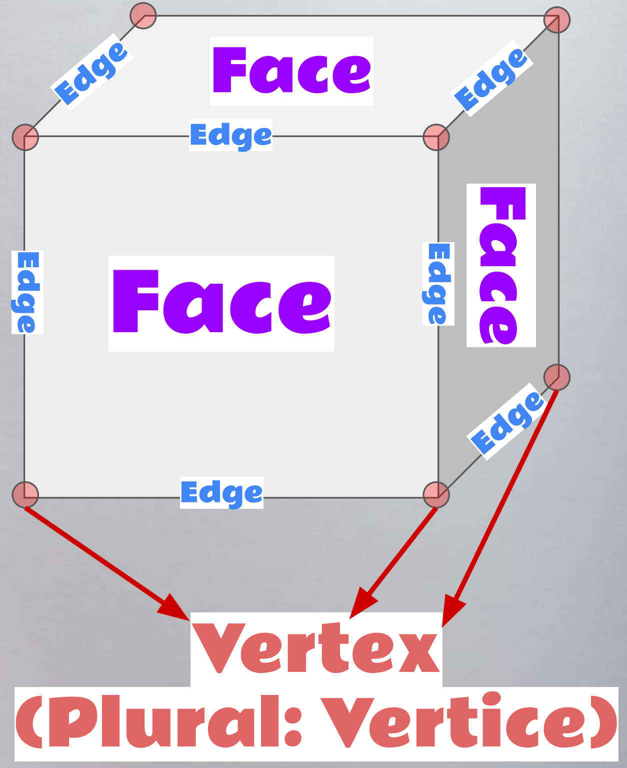
3x3x3=27 cubes



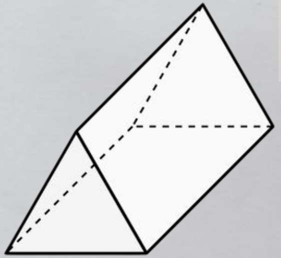
4x4x4=64 cubes

The number 8, 27, 64, 125... are perfect cubes!

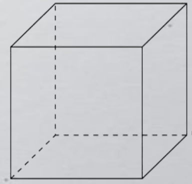
Note #35 3D Shapes



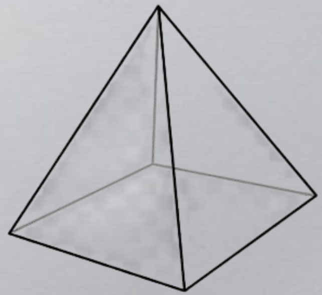
Rectangular Prism



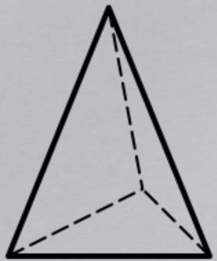
Triangular Prism



Cube(Square Prism)

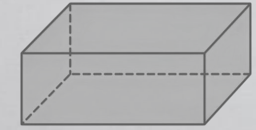


Square Pyramid

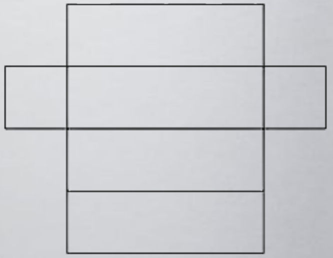


Triangular Pyramid

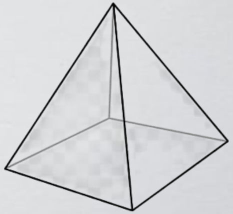
Note #36 Nets



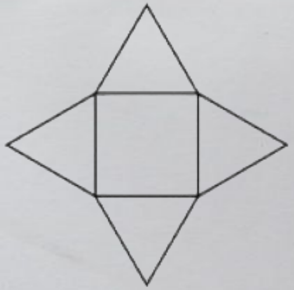
Rectangular Prism



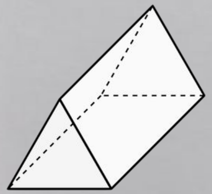
Nets



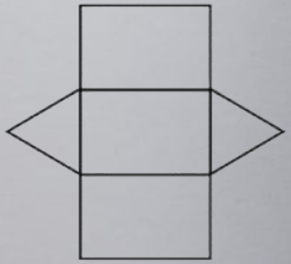
Square Pyramid



Nets

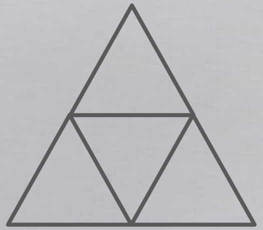


Triangular Prism



Nets

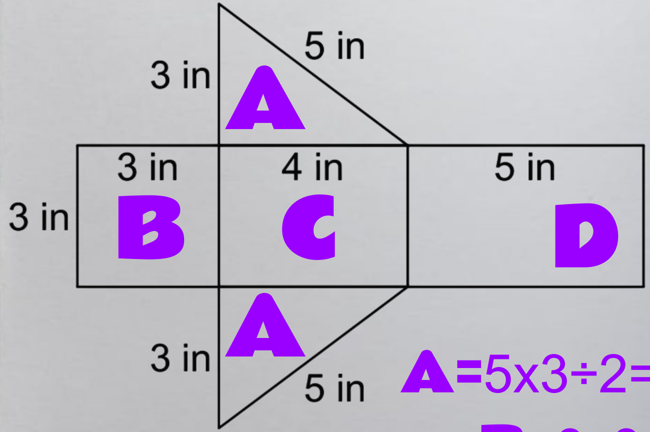
Triangular Pyramid



Nets

Note #46 Surface Area

To find the surface area, we need to find *the area of each face* from its net!

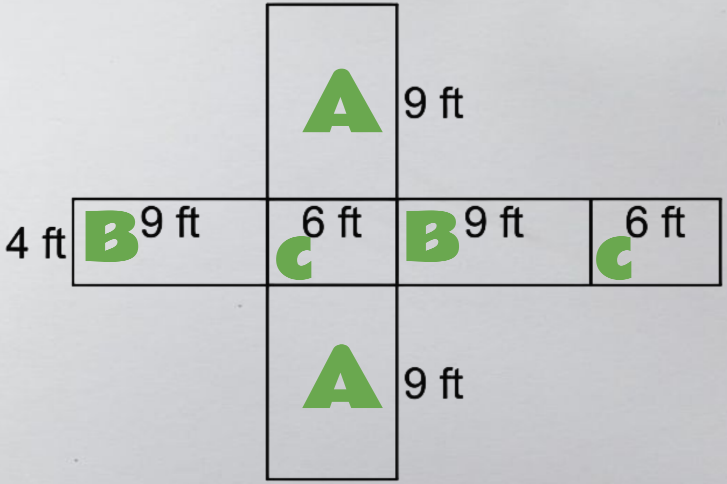


A=5x3÷2=7.5

B=3x3=9

C=3x4=12 D=3x5=15

Surface Area=
7.5+7.5+9+12+15= 51 in^2
(two triangles)



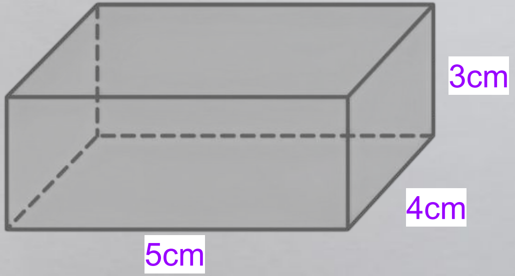
A=9x6=54 B=9x4=36

C=6x4=24

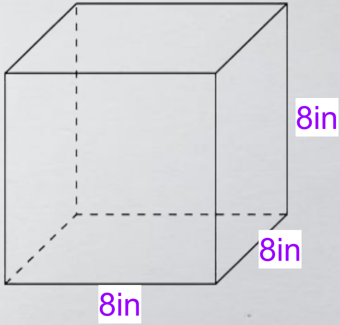
Surface Area=
54+54+36+36+24+24= 228 ft^2
(two same faces)

Note #57 Volume

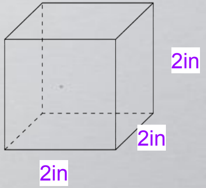
To find the volume, we need to find **the base area** and then **multiply** by the **height!**



Volume= $\underbrace{(5 \times 4)}_{\text{Base area}} \times \underbrace{3}_{\text{Height}} = 60\ cm^3$

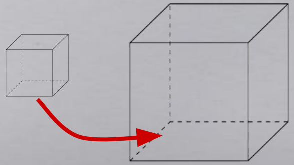


Volume=
$(8 \times 8) \times 8$
$= 512\ cm^3$



Volume=
$(2 \times 2) \times 2 = 8\ cm^3$

How many small cubes (2-inch side length) can fit in the big cube (8-inch side length)?



$512 \div 8$
$= 64$ cubes!